

COURSE TITLE : TYRE TECHNOLOGY
COURSE CODE : 6093
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/6
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Statistics ,design and development of pneumatic tyres	17
2	Bias and Radial tyres and tubes	21
3	Two wheeler tyres ,Auto tyres, Solid tyre and Re -treading	21
4	Tyre testing	16
TOTAL		75

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the statistics ,design and development of pneumatic tyres .
2	2	Comprehend the raw material ,components ,building and curing of bias and radial tyres and tubes.
3	3	Comprehend the raw material ,components ,building and curing of two wheeler tyres,auto tyres,solid tyre and re –treading.
4	4	To understand the various tyre testing methods.

SPECIFIC OUTCOME

MODULE I STATISTICS, DESIGN AND DEVELOPMENT OF PNEUMATIC TYRES

1.1.0 To understand the statistics, design and development of pneumatic tyres

- 1.1.1. Describe the history, invention and development of tyre.
- 1.1.2. Describe the statistics relating production, consumption,import and export of tyre products.
- 1.1.3. List out the major tyre manufacturing industries in India and abroad.
- 1.1.4. Define pneumatic tyre, Identify different components and functions of pneumatic tyre.
- 1.1.5. Describe the basic tyre constructions.
- 1.1.6. Distinguish between bias and radial tyre.
- 1.1.7. Define aspect ratio, general system indicating tyre size and dimensions.

- 1.1.8. Explain different components, method of production, raw materials, and their functions with a neat sketch.
- 1.1.9. Describe tyre reinforcements, requirements of tyre cords.
- 1.1.10. Explain advantages and disadvantages of cotton, rayon, nylon, polyester, glass, steel and aramid.
- 1.1.11. Describe the auxiliary materials.

MODULE II BIAS AND RADIAL TYRES AND TUBES

- 2.2.0 Comprehend the raw material ,components ,building and curing of bias and radial tyres and tubes.
- 2.2.1. Explain production of bead unit ,inner liner, carcass, sidewall, belt, breaker, tread and chaffer.
- 2.2.2. Design suitable compounds for components based on their final vulcanization properties.
- 2.2.3. State common tread patterns and its relevance to passenger, truck , and off- the -road tyres.
- 2.2.4. Describe the need of adhesive treatment for tyre cords and requirement of adhesive solution.
- 2.2.5. Describe adhesive treatment for rayon, nylon, and polyester.
- 2.2.6. Explain the calendaring of fabric and cutting in to carcass plies.
- 2.2.7. Explain the different parts of tyre building machine used for bias and radial tyre.
- 2.2.8. Explain the building process of bias and radial tyres.
- 2.2.9. Explain the curing of bias tyre and radial tyre using BOM press and autoform press.
- 2.2.10. Describe PCI, flat spotting and aquaplaning.
- 2.2.11. Explain the functions, properties, and suitable formulations for curing bag, curing bladder and diaphragm.
- 2.2.12. Describe the production process, typical compound, curing, inspection and functions of automobile inner tube.
- 2.2.13. Describe tubeless tyre.
- 2.2.14. Explain the production of tyre flap

MODULE III CYCLE TYRES,SOLID TYRE AND RE –TREADING

- 3.3.0 Comprehend the raw material ,components ,building and curing of cycle tyres, solid tyre and re –treading
- 3.3.1. Illustrate different components of cycle tyre.
- 3.3.2. Explain monoband and collapsible drum method for the production of cycle tyre.
- 3.3.3. Explain the production process of cycle tube.
- 3.3.4. Describe the curing of cycle tyre and tube.
- 3.3.5. Design typical compound for cycletyre and tube.
- 3.3.6. Describe the production of solid tyre with typical formulation, field of applications, and advantages and disadvantages.

- 3.3.7. Compare solid tyre with pneumatic tyre.
- 3.3.8. Explain the necessity and importance of retreading process.
- 3.3.9. Explain the retreading process by hot and cold process.
- 3.3.10. Identify the machinery used for retreading process.
- 3.3.11. Design typical compound for truck tyre tread, passenger car tyre tread, and cushion gum used in conventional and precured retreading process.
- 3.3.12. Compare hot and cold process of retreading

MODULE IV TYRE TESTING

4.4.0 To understand the various tyre testing methods

- 4.4.1. Explain laboratory testing of tyre cords and compound.
- 4.4.2. Describe laboratory testing of tyres.
- 4.4.3. Explain nondestructive tests such as dimensions, size factor, tread arc width, non skid depth, hardness, X-ray, IR, Ultrasonic, microwave, holography.
- 4.4.4. Explain destructive test on tyre like endurance test and plunger test.
- 4.4.5. Describe cut tyre analysis.
- 4.4.6. Describe highway test of tyres.
- 4.4.7. Explain destructive and nondestructive tests done on automobile inner tube.
- 4.4.8. Describe the procedure for the determination of cord strength and casing strength of cycle tyre.
- 4.4.9. Explain tension set of cycle tyre and tube.
- 4.4.10. Describe denison bend test.

CONTENT DETAILS

MODULE-I

Statistics, Design and development of pneumatic tyres

History, invention and development of tyres –bias, radial. Tubeless, tube – Statistics –production, consumption import and export of tyre products .Major tyre manufacturing industries in India and abroad ,Pneumatic tyres –definition –different components –functions . Different tyre construction – bias –radial . Aspect ratio –tyre size and dimension different components –method of production –raw material –Tyre reinforcement –requirement of tyre cord -Different cords –cotton –rayon –nylon – polyester –glass –steel –aramid. Auxiliary materials

MODULE-II

Bias ,Radial tyres and tubes

Production of Bead unit –inner liner –carcass –belt –breaker –tread –chaffer .compound design for each components. Tread patterns –lug, rib ,semi lug. Adhesive treatment for tyre cords -nylon-rayon and

polyester – adhesive formulation for each tyre cords .Calendaring of fabric and cutting into carcass plies .Tyre building machine –different parts –bias building machine and radial building machine .building process of bias and radial tyres .Curing of tyres –BOM press –auto form press .PCI –Flat spotting –Aqua planning .Curing bags –functions –properties –formulations .Automobile inner tube –production –typical compound – curing .Tubeless tyre – tyre flap .

MODULE-III

Cycle tyres , solid tyres and re- treading.

Cycle tyres–components –production–monoband and collapsible drum method – Cycle tube production and curing –compound . Solid tyres- formulations. Applications – advantages and disadvantages. compare solid tyre and pneumatic tyre .Retreading process –hot and cold process –machinery. compound design for truck tyre tread, passenger car tread ,cushion gum for conventional and procured retreading process .comparison .

MODULE-IV

Tyre testing

Testing of tyre cords and compounds – testing of tyres –non destructive tests -dimension ,size factor, tread arc width ,non skid depth , hardness, X-ray, IR, Ultra sonic, micro wave , holography , – destructive tests –endurance tests and plunger tests . Cut tyre analysis –High way test of tyre. Destructive and non destructive tests of inner tubes .Tension set, cord strength, casing strength of cycle tyre and tube – Denison bend test.

REFERENCE BOOKS

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| 1. Tyre Technology | -F.J.Kovac (Publisher - Goodyear Tyre & Rubber Company 4 th Edition) |
| 2. Rubber Technology and Manufacture | - C.M.Blow (Publisher – Butterworth Hainemann Ltd 2 nd Edition) |
| 3. Vanderbilt Rubber Hand book | -R.T Vanderbilt (Publisher -R.T Vanderbilt,13 th Edition |
| 4. Rubber Technology and Manufacture | – Published by SBP |
| 5. IS specifications. | |